

SRS Project Assignments

Software Requirements Specification
CSE Course | Academic Year 2024-2025

Each student is assigned one real-world software project. The task is to compile a complete Software Requirements Specification (SRS) document based on the assigned project. Carefully read the project description, understand the system, and document all functional and non-functional requirements following the IEEE 830 SRS standard.

#	Student ID	Student Name	Project Title
1	12411002	Md. Zehadul Islam	bKash Mobile Financial Service
2	12411003	Ashfaq Ahamed Siam	Nagad Digital Financial Platform
3	12411004	Junaid Islam Rakin	Daraz E-Commerce Marketplace
4	12411005	Esha Sanjida	Shajgoj Beauty & Lifestyle Platform
5	12411006	Rasdid Islam Alif	MyGP Grameenphone Self-Service App
6	12411007	Fahidul Islam Taspi	Robi MyRobi Customer App
7	12411011	Md Saad Khan	University Central Academic Portal
8	12411012	Mohannad Arafat	Online Exam Control and Result Management System
9	12411013	Meherab Hossain	Digital Course Material and LMS Portal
10	12411016	Aongon Saha	YouTube-Like Video Streaming Platform
11	12411017	Sawrav Howlader	Facebook Social Network Platform
12	12411021	Dilara hossen Dola	Pathao Ride-Sharing and Delivery App
13	12411022	Md Imam Hossain	Shohoz Bus Ticketing and Event Platform
14	12411023	Navila Binte Anower	Chaldal Online Grocery Delivery
15	12411024	Imam Hasan Hridoy	10 Minute School Online Learning Platform
16	12411025	Md. Tangimush Shifat	Bikroy.com Classified Ads Marketplace
17	12411026	Fatima Khatun Tammi	GPay / SSL Commerz Payment Gateway
18	12411027	Protik Sutar	Prothom Alo News Portal
19	12411028	Sanjida Islam Richi	Bangladesh Railway E-Ticketing System
20	12411029	Joyeta Das	Instagram-Like Photo and Reels Sharing App
21	12411030	Arafat Abid	WhatsApp Messaging and Calling Application
22	12411031	Md Hasib	Google Maps Navigation and Location Service
23	12411033	Samia Islam Arim	Rokomari Online Bookstore
24	12411035	Tuanur Tisa	Spotify Music Streaming Service
25	12411037	Omor Faruk Siam	Uber Ride-Hailing Application

26	12411039	Anita Sultana Flora	Foodpanda Food Delivery Platform
27	12411040	Tousif Rahman Mihir	Airbnb Short-Term Rental Marketplace
28	12411041	Sayma Alam Esa	LinkedIn Professional Networking Platform
29	12411042	Md. Rifat Hossain	Netflix Video on Demand Streaming Platform
30	12411043	Sourav Biswas	Hospital Appointment and Patient Management System
31	12411046	Sifat Ullah	Amazon E-Commerce and Fulfillment Platform
32	12411047	Lamia Akhter Moon	TikTok Short Video Platform
33	12411050	Marzan Akter	Google Search Engine
34	12411052	Indronil Howlader Akash	Zoom Video Conferencing Platform
35	12411055	Khalid Hasan	GitHub Code Repository and Collaboration Platform
36	12411058	Md Fahim Ahmed	Twitter / X Social Microblogging Platform
37	12411061	Md. Emon Howlader	Blood Donor Management System
38	12411062	Sumaiya Khan	Canva Online Graphic Design Tool
39	12411063	Amrin Hossain	Online Banking and Mobile Banking Portal
40	12411066	Md Jamil Abedin Naim	Weather Forecasting and Alert Application
41	12411068	Fatima Jannat Dola	Job Portal and Recruitment Platform
42	12321076	Muhammad Imtiaz Ahmed	Telemedicine and Online Doctor Consultation App
43	12421106	Md Touhidul Islam	Smart City Waste Management System
44	12321106	Bappy Barai	E-Government Citizen Service Portal
45	12321027	Miadam mahathir billah	Reddit-Like Community Discussion Platform

Project 01 bKash Mobile Financial Service

Assigned To:	Md. Zehadul Islam	Student ID:	12411002
--------------	-------------------	-------------	----------

Project Overview

bKash is Bangladesh's most widely used mobile financial service platform, allowing millions of users to send and receive money, pay bills, recharge mobile credit, and make online purchases directly from their mobile phones. It serves both banked and unbanked populations across urban and rural areas, making financial access available to people who never had a bank account.

Your task is to analyze the existing bKash system and prepare a complete Software Requirements Specification document. You will identify all major actors such as customers, agents, merchants, and admins, and document every functional module including registration, transaction flow, balance inquiry, and security features. The SRS must reflect how the real system behaves and what users actually need.

System Constraints

- Must support both Android and iOS platforms
- Transaction security must follow Bangladesh Bank guidelines
- System must handle minimum 10,000 concurrent users
- OTP-based two-factor authentication is mandatory
- No offline mode; real-time transaction processing required

Project Duration	12 weeks	Estimated Budget	BDT 25,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 02

Nagad Digital Financial Platform

Assigned To:	Ashfaq Ahamed Siam	Student ID:	12411003
--------------	--------------------	-------------	----------

Project Overview

Nagad is a government-backed mobile financial service operated by Bangladesh Post Office in partnership with Third Wave Technologies. It offers cash-in, cash-out, send money, mobile recharge, and utility bill payment services. Nagad gained rapid popularity due to its lower transaction fees and easy onboarding process using National ID-based verification without visiting a branch.

You will document the complete Nagad system from a software requirements perspective. Your SRS must cover the agent management system, customer-facing mobile app, and the backend administration panel. Pay close attention to the NID-based e-KYC process, transaction limits, and the notification system. Also document non-functional requirements like performance, availability, and regulatory compliance.

System Constraints

- e-KYC must integrate with the national NID database
- Daily transaction limits must comply with Bangladesh Bank rules
- System uptime must be at least 99.5 percent
- SMS and push notification support required
- Must work on low-end Android devices with 2G connectivity

Project Duration	12 weeks	Estimated Budget	BDT 20,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 03 Daraz E-Commerce Marketplace

Assigned To:	Junaid Islam Rakin	Student ID:	12411004
--------------	--------------------	-------------	----------

Project Overview

Daraz is South Asia's largest e-commerce platform operating in Bangladesh, Pakistan, Sri Lanka, Nepal, and Myanmar. It connects millions of buyers with thousands of sellers, offering products across categories like electronics, fashion, groceries, and home goods. The platform runs major sale events like 11.11 and features flash sales, vouchers, and installment-based payment options.

Your SRS must cover the full lifecycle of a Daraz transaction from product listing to delivery confirmation. Document the seller center module, buyer-facing shopping app, logistics and tracking integration, return and refund process, and the review system. You should also identify the recommendation engine requirements and how the search and filter system should behave for millions of SKUs.

System Constraints

- Product catalog must support millions of SKUs with fast search
- Payment gateway must support bKash, Nagad, cards, and COD
- Seller onboarding must include identity and trade license verification
- Return window must be configurable per product category
- Mobile app must function on Android 6.0 and above

Project Duration	14 weeks	Estimated Budget	BDT 40,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 04

Shajgoj Beauty & Lifestyle Platform

Assigned To:	Esha Sanjida	Student ID:	12411005
--------------	--------------	-------------	----------

Project Overview

Shajgoj is Bangladesh's leading beauty and lifestyle e-commerce platform focused on cosmetics, skincare, haircare, and wellness products. It targets urban women who want access to both local and international brands in one place. The platform includes a content section with beauty tips, tutorials, and product reviews written by beauty enthusiasts, making it both a shopping and information destination.

Your SRS should focus on the product discovery experience, review and rating system, content management for beauty articles, and the loyalty points program. Document the requirements for the recommendation engine that suggests products based on skin type and beauty goals. Also cover the seller portal requirements and how brand pages are managed differently from general product listings.

System Constraints

- Product pages must include ingredient lists and usage instructions
- Review system must support image and video uploads
- Loyalty points must auto-expire after 12 months
- Content and shopping modules must be accessible from the same app
- Must support guest checkout without mandatory registration

Project Duration	10 weeks	Estimated Budget	BDT 15,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 05 MyGP Grameenphone Self-Service App

Assigned To:	Rasdid Islam Alif	Student ID:	12411006
--------------	-------------------	-------------	----------

Project Overview

MyGP is the official self-service mobile application of Grameenphone, Bangladesh's largest mobile operator. It lets customers check their balance, buy internet and call packages, view usage history, pay bills, activate or deactivate services, and contact support without calling a helpline. The app serves tens of millions of active users and handles real-time interactions with Grameenphone's core telecom infrastructure.

You will document the MyGP system covering the SIM-based auto-login mechanism, package browsing and purchase flow, usage dashboard, bill payment integration, and the referral and reward system. Pay special attention to how the app interacts with the telecom backend in real time and what happens when a user switches between prepaid and postpaid accounts. Document all edge cases like low balance warnings and service deactivation flows.

System Constraints

- SIM-based authentication must work without internet on first login
- Data usage display must refresh every 5 minutes maximum
- Package activation must reflect within 60 seconds of purchase
- App must support both prepaid and postpaid account types
- Must work on Android 5.0 and iOS 12 and above

Project Duration	10 weeks	Estimated Budget	BDT 18,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 06 Robi MyRobi Customer App

Assigned To:	Fahidul Islam Taspi	Student ID:	12411007
--------------	---------------------	-------------	----------

Project Overview

MyRobi is the self-care application for Robi Axiata, the second largest mobile operator in Bangladesh. It provides subscribers with tools to manage their account, track data and minutes usage, subscribe to offers, pay bills, and access customer support through chat and callback features. The app also features a loyalty program called Robi Points where users earn and redeem rewards for using Robi services.

Your SRS should document the account management module, offer discovery and subscription flow, detailed usage reports with graphical breakdowns, and the Robi Points reward system. You also need to cover the chatbot-based support system and how it escalates to a live agent. Document the notification system that alerts users about data consumption, offer expiry, and promotional campaigns.

System Constraints

- Loyalty points must sync in real time with transaction events
- Chatbot must handle at least 50 predefined intents before escalation
- Usage graphs must display daily breakdowns for the last 30 days
- Offer subscription must send confirmation SMS immediately
- App must support Bangla and English language toggling

Project Duration	10 weeks	Estimated Budget	BDT 16,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 07 University Central Academic Portal

Assigned To:	Md Saad Khan	Student ID:	12411011
--------------	--------------	-------------	----------

Project Overview

A University Academic Portal is the central web-based system used by students, faculty, and administrative staff to manage all academic activities. It handles student enrollment, course registration, grade submission, transcript generation, attendance tracking, and library integration. Universities like BUET, DU, and private universities in Bangladesh rely on such systems to streamline communication between all academic stakeholders.

Your SRS must cover the full academic management cycle starting from student admission to graduation. Document the course registration system with credit hour validation, the grade submission workflow for faculty, the academic calendar management, and the student transcript generation module. Also address role-based access control where students, teachers, advisors, and admins each have different permissions and views.

System Constraints

- Role-based access control must be enforced at all module levels
- Grade submission must be locked after the official deadline
- Course registration must enforce prerequisite and credit limit rules
- Transcript PDF generation must include digital signature
- System must support at least 20,000 active student accounts

Project Duration	16 weeks	Estimated Budget	BDT 30,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 08 Online Exam Control and Result Management System

Assigned To:	Mohannad Arafat	Student ID:	12411012
--------------	-----------------	-------------	----------

Project Overview

An Exam Control System is used by universities and education boards to manage the full examination process from scheduling to result publication. It handles seat plan generation, hall allocation, invigilator assignment, answer script tracking, mark entry, result processing, and grade card distribution. Bangladesh's National University and many private universities depend on such systems to handle tens of thousands of students per semester.

You will document all workflows involved in running an examination including pre-exam setup, during-exam management, and post-exam result processing. Your SRS should cover the mark entry system with moderation and re-check request handling, the result calculation engine with GPA computation, and the result publication portal. Also document the tabulation sheet generation and the process for handling incomplete or withheld results.

System Constraints

- Mark entry must support double-entry verification to prevent errors
- Result publication must be triggered manually by authorized admins only
- System must generate seat plans automatically from enrollment data
- Re-check request portal must be available for limited period after result
- All result data must be archived and retrievable for 10 years

Project Duration	14 weeks	Estimated Budget	BDT 22,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 09 Digital Course Material and LMS Portal

Assigned To:	Meherab Hossain	Student ID:	12411013
--------------	-----------------	-------------	----------

Project Overview

A Learning Management System (LMS) is an online platform where teachers upload course materials, assignments, and resources while students access them, submit assignments, and track their learning progress. Platforms like Google Classroom, Moodle, and local university portals serve this purpose across Bangladesh. These systems became critical during COVID-19 and remain essential for blended learning environments.

Your SRS must document the course creation and management module, assignment submission and grading workflow, discussion forum system, and progress tracking dashboard. Cover the file upload requirements for different material types including PDFs, videos, and slides. Also document the notification system that alerts students about new materials, assignment deadlines, and graded submissions.

System Constraints

- File upload must support PDF, DOCX, PPT, MP4, and ZIP formats
- Assignment submission must record timestamp and prevent late uploads
- Discussion forum must support threaded replies and file attachments
- Progress dashboard must show completion percentage per course
- System must support at least 500 concurrent users during peak hours

Project Duration	12 weeks	Estimated Budget	BDT 18,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 10

YouTube-Like Video Streaming Platform

Assigned To:	Aongon Saha	Student ID:	12411016
--------------	-------------	-------------	----------

Project Overview

YouTube is the world's largest video platform where creators upload content and viewers watch, like, comment, and subscribe to channels. It handles billions of video views daily using adaptive streaming technology that adjusts video quality based on internet speed. In Bangladesh, YouTube is used heavily for education, entertainment, news, and by a growing creator economy that earns through the YouTube Partner Program.

Your SRS should document the video upload and processing pipeline, channel management system, recommendation algorithm requirements, monetization module, and the comment and moderation system. Cover the requirements for adaptive video streaming across different resolutions from 144p to 4K. Also document the subscription notification system, the analytics dashboard for creators, and the copyright content management requirements.

System Constraints

- Video processing must support MP4, AVI, MOV, and MKV upload formats
- Adaptive bitrate streaming must be supported for variable connections
- Content moderation must flag potentially harmful content automatically
- Creator analytics must update with maximum 24-hour delay
- Video player must support picture-in-picture and background play

Project Duration	20 weeks	Estimated Budget	BDT 80,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 11 Facebook Social Network Platform

Assigned To: **Sawrav Howlader**

Student ID: **12411017**

Project Overview

Facebook is the world's most used social network with nearly three billion active users. It allows people to share posts, photos, and videos, connect with friends and family, join communities through Groups, follow Pages, and interact through reactions, comments, and Messenger. In Bangladesh, Facebook is the dominant social media platform used for personal communication, news consumption, business promotion, and community organizing.

Your SRS must cover the news feed algorithm requirements, post creation and media upload module, friend and follow system, Groups and Pages management, and the Messenger integration. Document the privacy settings system that lets users control who sees their content. Also cover the advertising system requirements for business accounts, the notification engine, and the content reporting and moderation workflow.

System Constraints

- News feed must rank content based on engagement and relevance signals
- Privacy settings must be enforceable at post, account, and page level
- Messenger must support text, voice note, image, and video messages
- Business pages must have separate analytics from personal profiles
- Reported content must be reviewed within 24 hours of submission

Project Duration	20 weeks	Estimated Budget	BDT 1,00,00,000
-------------------------	----------	-------------------------	-----------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 12

Pathao Ride-Sharing and Delivery App

Assigned To:	Dilara hossen Dola	Student ID:	12411021
--------------	--------------------	-------------	----------

Project Overview

Pathao is Bangladesh's homegrown super-app that started as a motorcycle ride-sharing service and expanded into car rides, food delivery, and courier services. It connects riders and drivers in real time using GPS-based matching, dynamic pricing, and route optimization. The platform operates across Dhaka, Chittagong, and other major cities and has become a daily necessity for urban commuters and food lovers.

Your SRS must cover the real-time driver-rider matching system, GPS tracking and route management, dynamic fare calculation, trip history and rating system, and the food delivery module including restaurant onboarding and order management. Document the payment integration with mobile banking and cash options. Also cover the driver registration and verification workflow and the admin panel for monitoring active trips.

System Constraints

- GPS location must update every 5 seconds during active trips
- Driver matching must complete within 30 seconds of request
- Surge pricing algorithm must reflect real-time demand and supply
- Driver onboarding must verify NID and vehicle documents
- Trip cancellation policy must trigger automatic penalty calculations

Project Duration	16 weeks	Estimated Budget	BDT 50,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 13 Shohoz Bus Ticketing and Event Platform

Assigned To:	Md Imam Hossain	Student ID:	12411022
--------------	-----------------	-------------	----------

Project Overview

Shohoz is Bangladesh's leading online ticketing platform for intercity bus travel, launches, and events. It allows users to search routes, compare bus operators, select seats on an interactive seat map, and pay online for confirmed tickets. The platform has removed the need to stand in long queues at bus terminals and made travel planning accessible from a smartphone for millions of Bangladeshis.

Your SRS should document the route and schedule management system, interactive seat selection module, booking and ticketing workflow, payment processing, and the cancellation and refund policy engine. Cover the operator management portal where bus companies manage their fleet, schedules, and pricing. Also document the QR-code-based boarding pass system and the notification flow that sends booking confirmations and trip reminders.

System Constraints

- Seat inventory must update in real time to prevent double booking
- QR code tickets must be verifiable offline by bus staff
- Cancellation refund must auto-process within 48 hours
- Payment must support bKash, Nagad, Rocket, and card payments
- Search results must appear within 3 seconds for any route query

Project Duration	12 weeks	Estimated Budget	BDT 20,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 14 Chaldal Online Grocery Delivery

Assigned To:	Navila Binte Anower	Student ID:	12411023
--------------	---------------------	-------------	----------

Project Overview

Chaldal is Bangladesh's most popular online grocery delivery service offering fresh vegetables, fruits, meat, dairy, household essentials, and packaged goods delivered within 1 to 2 hours. Operating primarily in Dhaka, it maintains multiple dark stores across the city to enable hyper-local fast delivery. Chaldal has changed how urban families shop for daily necessities and set the benchmark for quick commerce in Bangladesh.

Your SRS must cover the product catalog and inventory management system, order placement and slot selection workflow, real-time delivery tracking, warehouse picking and packing module, and the customer subscription plan for regular buyers. Document the freshness guarantee system and how product returns are handled. Also cover the dark store management requirements for inventory replenishment and expiry tracking.

System Constraints

- Inventory must sync across all dark stores in near real time
- Delivery slot availability must reflect actual rider capacity
- Perishable item orders must be prioritized in picking sequence
- Return requests for fresh items must be processed within 2 hours
- App must support scheduled delivery up to 3 days in advance

Project Duration	14 weeks	Estimated Budget	BDT 28,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 15 10 Minute School Online Learning Platform

Assigned To:	Imam Hasan Hridoy	Student ID:	12411024
--------------	-------------------	-------------	----------

Project Overview

10 Minute School is Bangladesh's largest ed-tech platform offering video lessons, live classes, quizzes, and full course content for students from class 6 through university admission and professional skill development. Founded by Ayman Sadiq, the platform has democratized access to quality education for students across Bangladesh regardless of their location or economic background.

Your SRS should document the course enrollment and progress tracking system, live class module with interactive Q&A, quiz and assessment engine with automated grading, and the certificate generation system. Cover the content creator portal where teachers upload and manage lessons. Also document the student performance analytics dashboard and the mobile-first experience requirements since most users access from low-end smartphones with limited data.

System Constraints

- Video lessons must be downloadable for offline viewing
- Quiz engine must support MCQ, short answer, and fill-in-the-blank types
- Live classes must support minimum 5,000 concurrent viewers
- Certificate generation must include QR verification code
- App must work efficiently on 2G/3G connections for rural users

Project Duration	14 weeks	Estimated Budget	BDT 25,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 16 Bikroy.com Classified Ads Marketplace

Assigned To:	Md. Tangimush Shifat	Student ID:	12411025
--------------	----------------------	-------------	----------

Project Overview

Bikroy is Bangladesh's biggest online classifieds marketplace where individuals and businesses post ads to buy and sell products, vehicles, real estate, jobs, and services. It functions like a digital notice board connecting buyers and sellers without the platform being involved in the transaction itself. Users can post free ads or pay for featured placement to get more visibility for their listings.

Your SRS must document the ad creation and management workflow, category and subcategory system, search and filter engine, ad promotion and featured listing module, and the messaging system between buyers and sellers. Cover the verification system for high-value categories like cars and real estate. Also document the admin panel for ad moderation, fraud detection requirements, and the reporting mechanism for suspicious listings.

System Constraints

- Free ad posting must be limited to 5 active ads per unverified account
- Search must support location-based filtering by district and upazila
- Photo upload per ad must support minimum 10 images
- Promoted ads must expire automatically after paid duration ends
- Fraud report must notify admin within 1 hour of submission

Project Duration	12 weeks	Estimated Budget	BDT 18,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 17 GPay / SSL Commerz Payment Gateway

Assigned To:	Fatima Khatun Tammi	Student ID:	12411026
--------------	---------------------	-------------	----------

Project Overview

SSLCommerz is Bangladesh's largest payment gateway connecting online businesses with all major payment methods including mobile banking (bKash, Nagad, Rocket), debit and credit cards (Visa, Mastercard), internet banking, and EMI options. Thousands of e-commerce and service businesses integrate SSLCommerz to accept payments from customers without building separate integrations for each payment method.

Your SRS should document the merchant onboarding and API integration system, the payment session initiation and callback handling workflow, refund processing module, and the transaction reporting dashboard for merchants. Cover the security requirements including SSL encryption, tokenization, and fraud detection. Also document the reconciliation system that matches payments received against orders placed and handles failed or pending transactions.

System Constraints

- Payment session must expire within 30 minutes of initiation
- All transactions must be encrypted using TLS 1.2 or higher
- Refund processing must complete within 5 business days
- Merchant dashboard must show real-time transaction status
- System must handle 500 transactions per second during peak events

Project Duration	14 weeks	Estimated Budget	BDT 35,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 18 Prothom Alo News Portal

Assigned To:	Protik Sutar	Student ID:	12411027
--------------	--------------	-------------	----------

Project Overview

Prothom Alo is Bangladesh's most widely read Bangla newspaper with a major digital presence through its website and mobile app. It publishes breaking news, in-depth analysis, sports coverage, entertainment content, and opinion pieces. The digital platform serves millions of readers daily and operates a subscription model for premium content alongside free ad-supported access for general readers.

Your SRS must document the content management system for journalists and editors, the article publishing workflow with editorial approval stages, the reader-facing website and app experience, and the subscription and paywall management module. Cover the push notification system for breaking news, the search and archive system for historical articles, and the advertisement placement and management requirements for the digital platform.

System Constraints

- Breaking news articles must be publishable within 5 minutes of approval
- Paywall must allow 5 free articles per month for non-subscribers
- Search must index and retrieve articles published within the last 20 years
- Push notifications must reach 90 percent of subscribed users within 2 minutes
- System must support both Bangla Unicode and English content

Project Duration	12 weeks	Estimated Budget	BDT 20,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 19 Bangladesh Railway E-Ticketing System

Assigned To: **Sanjida Islam Richi**

Student ID: **12411028**

Project Overview

Bangladesh Railway's e-ticketing system allows passengers to search train schedules, check seat availability, book tickets online, and print or download e-tickets for intercity travel. The system replaced the painful process of standing in long physical queues at railway stations and made ticket booking accessible from home. It handles peak demand during Eid seasons when millions of tickets are booked within hours of release.

You will document the train schedule and seat inventory management system, the user registration and booking workflow, the seat class selection module for different cabin types like Shobhon, Shulov, and AC berths, and the payment integration. Cover the Eid surge capacity requirements and the queue management system for high-demand releases. Also document the ticket cancellation and refund policy and the passenger verification system at boarding.

System Constraints

- Seat booking must be confirmed within 10 seconds and seats held for 15 minutes during payment
- System must handle 50,000 concurrent users during Eid ticket release
- E-ticket QR code must be scannable by railway staff without internet
- Cancellation refund policy must vary by time before departure
- Waiting list must automatically confirm when a booked seat is cancelled

Project Duration

16 weeks

Estimated Budget

BDT 35,00,000

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 20

Instagram-Like Photo and Reels Sharing App

Assigned To:	Joyeta Das	Student ID:	12411029
--------------	------------	-------------	----------

Project Overview

Instagram is a visual social media platform owned by Meta where users share photos, short videos (Reels), and Stories that disappear after 24 hours. It features a discovery engine that recommends content from accounts users don't follow, a direct messaging system, and shopping features for brands. In Bangladesh, Instagram is especially popular among youth and content creators in the fashion, food, and lifestyle space.

Your SRS must document the photo and video upload pipeline, Stories creation and viewing system with expiry logic, Reels algorithm and recommendation requirements, the explore discovery engine, and the DM system with media sharing. Cover the creator tools including analytics dashboard and business account features. Also document the hashtag system, tagged content management, and the content moderation requirements for community guidelines enforcement.

System Constraints

- Stories must automatically expire 24 hours after posting
- Reels must support video lengths from 15 seconds to 3 minutes
- Photo upload must support simultaneous posting of up to 10 images
- Explore algorithm must personalize results based on engagement history
- Business accounts must have access to audience demographics and reach data

Project Duration	18 weeks	Estimated Budget	BDT 70,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 21 WhatsApp Messaging and Calling Application

Assigned To:	Arafat Abid	Student ID:	12411030
--------------	-------------	-------------	----------

Project Overview

WhatsApp is the world's most used messaging application with over two billion users globally. It allows users to send text messages, voice notes, images, documents, and make voice and video calls over the internet. WhatsApp features end-to-end encryption for all messages and calls, group chats, broadcast lists, and WhatsApp Business for small businesses to communicate with customers at scale.

Your SRS should document the message delivery and read receipt system, media upload and compression pipeline, voice and video call quality requirements, group management with admin controls, and the end-to-end encryption architecture requirements. Cover the WhatsApp Business module including catalog management, quick replies, and automated greeting messages. Also document the message backup and restore system and the account deletion and data portability requirements.

System Constraints

- All messages must be end-to-end encrypted with no server-side readable content
- Message delivery confirmation must use three-state tick system
- Group size must support up to 1,024 members
- Voice call must maintain quality at minimum 64kbps bandwidth
- Media files must auto-download only on WiFi unless changed in settings

Project Duration	18 weeks	Estimated Budget	BDT 90,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 22 Google Maps Navigation and Location Service

Assigned To:	Md Hasib	Student ID:	12411031
--------------	----------	-------------	----------

Project Overview

Google Maps is the world's most widely used mapping and navigation application providing turn-by-turn directions, real-time traffic information, public transit routes, street view, and business location information. In Bangladesh it is used daily by commuters, delivery riders, and travelers to navigate Dhaka's complex road network and find restaurants, hospitals, ATMs, and other points of interest.

Your SRS must document the map rendering and tile loading system, real-time traffic data integration, route calculation engine for different transport modes, place search and autocomplete module, and the business listing and review system. Cover the offline map download requirements for areas with poor connectivity. Also document the location sharing feature, the contribution system where users report road changes, and the integration requirements for third-party apps using the Maps API.

System Constraints

- Map tiles must load within 2 seconds on a 3G connection
- Route recalculation must trigger within 10 seconds of a missed turn
- Offline map downloads must support area-based selection by city
- Place search must return results within 1 second with autocomplete
- User-submitted map edits must go through moderation before going live

Project Duration	20 weeks	Estimated Budget	BDT 1,20,00,000
------------------	----------	------------------	-----------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 23 Rokomari Online Bookstore

Assigned To:	Samia Islam Arim	Student ID:	12411033
--------------	------------------	-------------	----------

Project Overview

Rokomari is Bangladesh's largest online bookstore selling Bangla and English books across categories including academic textbooks, fiction, religious books, children's books, and international publications. It also sells stationery, electronics, and gift items. Rokomari has built a loyal customer base of book lovers and students who rely on it for finding books that are often unavailable in local markets.

Your SRS must document the book catalog management system with ISBN and author-based search, the shopping cart and wishlist module, order and delivery tracking, and the reader review and rating system. Cover the pre-order functionality for upcoming book releases and the out-of-stock notification system. Also document the publisher and author portal requirements and the promotional campaign management for sale events during Ekushey Book Fair and other seasonal events.

System Constraints

- Search must support both Bangla and English title and author queries
- Wishlist items must trigger email notification when price drops
- Out-of-stock books must show estimated restock date when available
- Pre-order payments must only be charged after book is in stock
- Order tracking must show warehouse, dispatch, and delivery stages

Project Duration	12 weeks	Estimated Budget	BDT 16,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 24

Spotify Music Streaming Service

Assigned To:	Tuanur Tisa	Student ID:	12411035
--------------	-------------	-------------	----------

Project Overview

Spotify is the world's most popular music streaming platform with over 600 million users globally. It offers access to millions of songs, podcasts, and audiobooks with personalized playlist recommendations powered by AI. Users can stream for free with ads or subscribe to Spotify Premium for offline listening, higher audio quality, and uninterrupted playback. Spotify is widely used in Bangladesh by the urban youth demographic.

Your SRS should document the music catalog and metadata management system, audio streaming pipeline with adaptive quality, playlist creation and sharing module, personalized recommendation engine requirements, and the offline download system. Cover the podcast and audiobook module with chapter navigation. Also document the artist dashboard for creators, the royalty calculation system requirements, and the social features like collaborative playlists and friend activity feed.

System Constraints

- Audio streaming must auto-adjust quality from 24kbps to 320kbps based on connection
- Offline downloads must be encrypted and tied to the account that downloaded them
- Recommendation engine must update Discover Weekly every Monday automatically
- Free tier must include audio advertisements every 3 to 4 songs
- Lyrics display must be synchronized with audio playback in real time

Project Duration	18 weeks	Estimated Budget	BDT 75,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 25 Uber Ride-Hailing Application

Assigned To:	Omor Faruk Siam	Student ID:	12411037
--------------	-----------------	-------------	----------

Project Overview

Uber is a global ride-hailing platform operating in Bangladesh that connects passengers with car and motorcycle drivers through a GPS-based mobile app. It features upfront pricing, cashless payment, driver ratings, and real-time trip tracking. Uber Bangladesh operates UberX, Uber Moto, and UberXL services and serves as a safer and more transparent alternative to traditional CNG auto-rickshaws and taxis.

Your SRS must document the ride request and driver matching algorithm requirements, real-time GPS tracking system, fare estimation and surge pricing engine, cashless payment integration, and the driver and rider rating system. Cover the driver onboarding and document verification module. Also document the trip safety features including the emergency SOS button, share trip functionality, and the incident reporting system for both drivers and riders.

System Constraints

- Driver matching must complete within 30 seconds of trip request
- Surge pricing multiplier must be displayed clearly before booking confirmation
- Emergency SOS must notify emergency contacts and share live location
- Driver document verification must be completed before first trip
- Cancellation fees must apply after 5 minutes of driver acceptance

Project Duration	16 weeks	Estimated Budget	BDT 60,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 26 Foodpanda Food Delivery Platform

Assigned To:	Anita Sultana Flora	Student ID:	12411039
--------------	---------------------	-------------	----------

Project Overview

Foodpanda is one of the leading food delivery platforms in Bangladesh operating in Dhaka, Chittagong, and other major cities. It connects customers with hundreds of restaurants, cloud kitchens, and grocery stores. Users can browse menus, customize orders, track delivery in real time, and pay online or with cash on delivery. Foodpanda also operates its own delivery fleet called Pandamart for grocery and household item delivery.

Your SRS should document the restaurant onboarding and menu management system, customer order placement and customization workflow, real-time order tracking with GPS, delivery rider assignment and routing, and the payment and discount system. Cover the rating and review system for restaurants and delivery experience. Also document the restaurant partner dashboard showing order volume, revenue, and customer feedback analytics.

System Constraints

- Order confirmation must reach restaurant within 30 seconds of placement
- Delivery time estimate must update dynamically based on real rider location
- Restaurant menu must sync with the app within 5 minutes of any update
- Discount coupons must validate against usage limits before applying
- Order must be refundable if restaurant does not confirm within 10 minutes

Project Duration	14 weeks	Estimated Budget	BDT 35,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 27 Airbnb Short-Term Rental Marketplace

Assigned To:	Tousif Rahman Mihir	Student ID:	12411040
--------------	---------------------	-------------	----------

Project Overview

Airbnb is a global online marketplace where property owners list their homes, apartments, and rooms for short-term rental to travelers. Guests browse listings with photos, reviews, and pricing, then book directly through the platform with secure payment holding. Airbnb handles the entire transaction including damage protection, host insurance, and dispute resolution, making it a trusted platform for both hosts and guests.

Your SRS must document the listing creation and management system including photo upload, availability calendar, pricing rules, and house rules. Cover the guest search and filter experience, the instant and request booking flows, the secure payment and payout system, and the review system for both guests and hosts. Also document the Superhost recognition system, the dispute resolution workflow, and the identity verification requirements for trust and safety.

System Constraints

- Host identity must be verified through government ID before listing goes live
- Payment must be held in escrow and released 24 hours after check-in
- Availability calendar must sync in real time to prevent double booking
- Reviews must be posted only after both guest and host review each other
- Cancellation policy must calculate refunds based on chosen policy type

Project Duration	18 weeks	Estimated Budget	BDT 80,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 28 LinkedIn Professional Networking Platform

Assigned To:	Sayma Alam Esa	Student ID:	12411041
--------------	----------------	-------------	----------

Project Overview

LinkedIn is the world's largest professional networking platform with over one billion members. It allows professionals to build their career profile, connect with colleagues and recruiters, share industry content, and apply for jobs. Companies use LinkedIn to post job openings, build employer brand, and find candidates. In Bangladesh, LinkedIn usage has grown rapidly among fresh graduates and professionals seeking career opportunities.

Your SRS should document the professional profile creation system including skills, endorsements, recommendations, and work history. Cover the job posting and application tracking system, the content feed and algorithm, the messaging and connection request workflow, and the recruiter search and filtering tools. Also document the LinkedIn Learning module integration, the company page management system, and the profile visibility and privacy control requirements.

System Constraints

- Profile completeness score must encourage users to fill all key sections
- Job alert must notify users within 1 hour of matching job posting
- Endorsements must only be allowed from verified first-degree connections
- Recruiter search must support filtering by skills, location, and experience
- Content algorithm must reduce engagement bait and prioritize professional content

Project Duration	18 weeks	Estimated Budget	BDT 65,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 29 Netflix Video on Demand Streaming Platform

Assigned To:	Md. Rifat Hossain	Student ID:	12411042
--------------	-------------------	-------------	----------

Project Overview

Netflix is the world's leading video streaming service with over 270 million paid subscribers globally. It offers a vast library of movies, TV series, documentaries, and original productions across multiple genres and languages. Users can stream on smart TVs, laptops, tablets, and smartphones with adaptive quality and multiple simultaneous streams depending on their subscription plan. Netflix in Bangladesh is accessed mainly through mobile and has local language content support.

Your SRS must document the content catalog and metadata system, the subscription plan management module, adaptive video streaming requirements, the recommendation engine, download for offline viewing, and multi-profile account management. Cover the content licensing and regional availability control system. Also document the parental control module, the continue watching feature, and the A/B testing framework requirements for UI and content experiments.

System Constraints

- Adaptive streaming must support resolutions from 480p to 4K HDR
- Each account must support up to 5 individual user profiles
- Offline downloads must expire within 30 days or 48 hours after first play
- Parental control PIN must block access to content above set rating
- Recommendation must update based on viewing history within 24 hours

Project Duration	20 weeks	Estimated Budget	BDT 1,00,00,000
------------------	----------	------------------	-----------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 30 Hospital Appointment and Patient Management System

Assigned To:	Sourav Biswas	Student ID:	12411043
--------------	---------------	-------------	----------

Project Overview

A Hospital Management System (HMS) is a software platform used by hospitals and clinics to manage patient records, doctor appointments, diagnostic reports, billing, pharmacy, and bed allocation. Major hospitals in Bangladesh like Square, United, and Labaid are moving toward digital HMS to reduce paperwork, prevent medical errors, and improve patient experience. The system connects doctors, nurses, lab technicians, and administrative staff on a single platform.

Your SRS must document the patient registration and Electronic Health Record (EHR) system, appointment booking and queue management module, doctor schedule and availability management, diagnostic report upload and access system, and the pharmacy and billing integration. Cover the ward and bed management module and the nurse station dashboard. Also document the patient portal requirements where patients can view reports, prescriptions, and upcoming appointment reminders.

System Constraints

- Patient records must be accessible only to treating doctors and authorized staff
- Appointment confirmation must be sent via SMS and app notification
- Diagnostic reports must be uploaded within 2 hours of completion
- Billing system must integrate with insurance claim processing
- System must comply with Bangladesh data protection and health data guidelines

Project Duration	16 weeks	Estimated Budget	BDT 45,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 31 Amazon E-Commerce and Fulfillment Platform

Assigned To:	Sifat Ullah	Student ID:	12411046
--------------	-------------	-------------	----------

Project Overview

Amazon is the world's largest e-commerce and cloud computing company. Its marketplace connects millions of third-party sellers with hundreds of millions of buyers globally. Amazon's Fulfilled by Amazon (FBA) service lets sellers store products in Amazon's warehouses and handles picking, packing, shipping, and returns. Prime membership provides subscribers with free fast shipping, streaming, and exclusive deals, creating a powerful loyalty ecosystem.

Your SRS should document the seller onboarding and product listing system, the search and ranking algorithm, the shopping cart and one-click ordering workflow, the FBA inventory and fulfillment module, and the Prime membership management system. Cover the customer review verification system and the A-to-Z guarantee for buyer protection. Also document the seller performance metrics dashboard and the advertising system for sponsored products and display ads.

System Constraints

- Product listing must include mandatory safety and compliance fields for regulated items
- Search ranking must account for relevance, price, seller rating, and Prime eligibility
- One-click ordering must complete in under 5 seconds with stored payment
- FBA inventory levels must trigger restocking alerts at configurable thresholds
- Seller account suspension must follow a documented warning and appeal process

Project Duration	20 weeks	Estimated Budget	BDT 1,50,00,000
------------------	----------	------------------	-----------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 32 TikTok Short Video Platform

Assigned To:	Lamia Akhter Moon	Student ID:	12411047
--------------	-------------------	-------------	----------

Project Overview

TikTok is a short-form video platform with over a billion active users where creators post videos ranging from 15 seconds to 10 minutes. Its For You Page (FYP) algorithm is considered the most powerful content recommendation engine in social media, capable of showing highly relevant content to new users within minutes of signup. TikTok is used in Bangladesh for entertainment, education, comedy, and increasingly for business promotion and influencer marketing.

Your SRS must document the video upload and processing pipeline, the FYP recommendation algorithm requirements, the video creation tools including filters, effects, and music library integration, the live streaming module, and the creator fund and monetization system. Cover the duet and stitch features for collaborative content. Also document the content moderation requirements, age verification for minors, and the TikTok Shop integration for in-video product linking and purchase.

System Constraints

- FYP must start personalizing within the first 3 videos watched by a new user
- Video upload must complete processing within 2 minutes for standard quality
- Live streaming must support minimum 10,000 concurrent viewers per channel
- Music used in videos must be cleared for copyright before publishing
- Content for users under 18 must have additional safety restrictions

Project Duration	20 weeks	Estimated Budget	BDT 1,20,00,000
------------------	----------	------------------	-----------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 33 Google Search Engine

Assigned To: **Marzan Akter**

Student ID: **12411050**

Project Overview

Google Search is the world's most used search engine processing over 8.5 billion searches per day. It crawls and indexes billions of web pages and returns ranked results in under a second based on hundreds of ranking signals. Google Search includes features like featured snippets, People Also Ask, image search, news panels, and knowledge graph cards that provide direct answers without needing to click a link.

Your SRS must document the web crawling and indexing pipeline requirements, the query parsing and intent recognition system, the ranking algorithm high-level requirements, the search result page layout and feature cards, and the auto-complete and spell correction module. Cover the image search and reverse image search requirements. Also document the Search Console for webmasters, the safe search content filtering system, and the localization requirements for showing location-relevant results.

System Constraints

- Search results must appear within 0.5 seconds of query submission
- Auto-complete must show relevant suggestions within 100 milliseconds of typing
- Safe search must be the default setting for users under 18
- Featured snippets must be extracted from indexed pages without manual curation
- Crawling must respect robots.txt rules and crawl rate limits

Project Duration 20 weeks

Estimated Budget BDT 2,00,00,000

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 34 Zoom Video Conferencing Platform

Assigned To:	Indronil Howlader Akash	Student ID:	12411052
--------------	-------------------------	-------------	----------

Project Overview

Zoom is the world's most popular video conferencing platform used for meetings, webinars, online classes, and virtual events. It surged to global prominence during COVID-19 and became a daily tool for students, professionals, and organizations. In Bangladesh, Zoom is widely used for remote work, university classes, government meetings, and international business communication. It supports up to 1,000 participants in a single meeting with screen sharing, recording, and breakout rooms.

Your SRS must document the meeting creation and scheduling system, the video and audio quality management and adaptive codec requirements, screen sharing and annotation module, breakout room management, cloud recording and transcript generation, and the waiting room and participant management controls. Cover the webinar module with different host, panelist, and attendee permission levels. Also document the Zoom App Marketplace integration requirements and the end-to-end encryption options for sensitive meetings.

System Constraints

- Meeting must support minimum 100 participants on the free plan
- Screen sharing must have less than 200 milliseconds of display latency
- Breakout rooms must allow host to broadcast messages to all rooms simultaneously
- Cloud recording must begin processing within 1 hour of meeting end
- Waiting room must allow host to admit or reject participants individually

Project Duration	16 weeks	Estimated Budget	BDT 60,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 35 GitHub Code Repository and Collaboration Platform

Assigned To: **Khalid Hasan**

Student ID: **12411055**

Project Overview

GitHub is the world's largest code hosting and collaboration platform used by over 100 million developers. It provides Git-based version control, pull request workflows, code review tools, issue tracking, project boards, and CI/CD automation through GitHub Actions. Students and professionals in Bangladesh use GitHub to store personal projects, collaborate on open-source software, and showcase their portfolio to potential employers.

Your SRS must document the repository creation and management system, the pull request and code review workflow, the issue and milestone tracking module, GitHub Actions automation pipeline requirements, and the security scanning and secret detection features. Cover the organization and team management system with permission levels. Also document the GitHub Pages hosting requirements, the API access and OAuth app integration system, and the starred repository and follow system for social discovery.

System Constraints

- Pull request must support inline code comments on specific lines
- Branch protection rules must block direct pushes to protected branches
- GitHub Actions workflow must trigger within 30 seconds of code push
- Repository must enforce 2FA for organization members on paid plans
- Fork and clone operations must complete within acceptable time for large repos

Project Duration

18 weeks

Estimated Budget

BDT 80,00,000

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 36

Twitter / X Social Microblogging Platform

Assigned To:	Md Fahim Ahmed	Student ID:	12411058
--------------	----------------	-------------	----------

Project Overview

Twitter, now rebranded as X, is a real-time social network where users post short messages called tweets, follow other accounts, retweet content, and engage through likes and replies. It is a major platform for news, political discourse, live event commentary, and public figure communication. In Bangladesh, Twitter/X is used by journalists, politicians, academics, and tech-savvy youth to follow global conversations and share opinions.

Your SRS should document the tweet creation system with character limits, media attachments, polls, and thread support, the timeline and algorithm-driven feed, the trending topics and hashtag system, the Lists and Circles features, and the X Premium subscription management module. Cover the notification system for mentions, replies, and direct messages. Also document the account verification system, the content moderation and appeal process, and the API access requirements for developers.

System Constraints

- Tweet character limit must be enforced at 280 for free and 25,000 for Premium
- Trending topics must update every 15 minutes based on real-time volume
- Lists must support both public and private visibility settings
- Direct message requests must require sender to follow or be approved
- API rate limits must be enforced per app token and per user token separately

Project Duration	18 weeks	Estimated Budget	BDT 90,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 37 Blood Donor Management System

Assigned To:	Md. Emon Howlader	Student ID:	12411061
--------------	-------------------	-------------	----------

Project Overview

A Blood Donor Management System connects people who need blood with willing donors in their area during medical emergencies. Platforms like Rokto.com.bd in Bangladesh allow patients or their families to post blood requests with blood group, location, and urgency, while registered donors receive notifications and can respond. The system has saved countless lives by reducing the time it takes to find compatible donors in critical situations.

Your SRS must document the donor registration and profile management system including blood group, donation history, and availability status, the blood request posting and matching algorithm, the notification and response system, and the donation history tracking with eligibility enforcement. Cover the location-based matching requirements to find donors nearest to the patient. Also document the hospital partnership module and the admin panel for managing requests, donor verification, and abuse prevention.

System Constraints

- Donor must be flagged ineligible for 3 months after each donation automatically
- Blood request must match and notify nearby eligible donors within 5 minutes
- Donor location must be shareable only after they agree to respond to a request
- Notification must be sent via SMS for users without internet access
- Admin must be able to verify and badge trusted regular donors

Project Duration	10 weeks	Estimated Budget	BDT 10,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 38 Canva Online Graphic Design Tool

Assigned To:	Sumaiya Khan	Student ID:	12411062
--------------	--------------	-------------	----------

Project Overview

Canva is an online design platform used by millions worldwide to create social media graphics, presentations, posters, flyers, resumes, and marketing materials without needing professional design skills. It offers a drag-and-drop editor with thousands of templates, stock photos, icons, and fonts. In Bangladesh, Canva is widely used by students, small business owners, content creators, and marketing teams to produce visually polished content quickly and affordably.

Your SRS must document the drag-and-drop design editor requirements, the template library and search system, the asset library management for photos, icons, and fonts, the collaborative editing and commenting module, and the design export options including PNG, PDF, MP4, and direct social media publishing. Cover the Brand Kit feature for teams to maintain consistent colors and fonts. Also document the version history system and the Canva for Teams shared workspace requirements.

System Constraints

- Editor must support real-time collaboration with visible cursor for each user
- Design export must complete within 30 seconds for standard resolution
- Template search must return relevant results within 1 second
- Brand Kit must enforce color palette and font restrictions for team members
- Version history must retain last 50 save states per design

Project Duration	16 weeks	Estimated Budget	BDT 55,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 39 Online Banking and Mobile Banking Portal

Assigned To:	Amrin Hossain	Student ID:	12411063
--------------	---------------	-------------	----------

Project Overview

An Online Banking Portal is a secure web and mobile platform provided by banks like Dutch Bangla Bank, BRAC Bank, and Islami Bank Bangladesh for customers to manage their accounts remotely. Customers can check balance, view transaction history, transfer funds, pay utility bills, open fixed deposits, apply for loans, and download bank statements without visiting a branch. Mobile banking apps have become essential for both personal and business banking customers across Bangladesh.

Your SRS must document the secure login and multi-factor authentication system, account dashboard and statement module, fund transfer workflow covering BEFTN and RTGS options, utility and merchant bill payment module, and the fixed deposit and loan application tracking system. Cover the card management module for blocking and managing debit and credit cards. Also document the audit logging requirements for all transactions and the customer service chat integration for banking support.

System Constraints

- All sessions must auto-logout after 10 minutes of inactivity
- Fund transfers above BDT 50,000 must require OTP re-verification
- Transaction history must be viewable for last 2 years minimum
- System must comply with Bangladesh Bank internet banking security guidelines
- Failed login attempts must lock account after 5 consecutive failures

Project Duration	16 weeks	Estimated Budget	BDT 50,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 40

Weather Forecasting and Alert Application

Assigned To:	Md Jamil Abedin Naim	Student ID:	12411066
--------------	----------------------	-------------	----------

Project Overview

A weather forecasting app provides real-time weather conditions, hourly and daily forecasts, severe weather alerts, and air quality information for any location. Apps like AccuWeather and the Bangladesh Meteorological Department's system are critical for farmers, fishermen, disaster management authorities, and everyday citizens planning outdoor activities. In Bangladesh, weather apps play a vital role during cyclone season and monsoon flooding events.

Your SRS must document the location detection and manual location management system, real-time weather data integration from meteorological APIs, hourly and 10-day forecast display requirements, severe weather alert push notification system, and the radar and satellite map module. Cover the air quality index display and health advisory system. Also document the Bangladesh-specific requirements for cyclone tracking, storm surge alerts, and flood warning integration with the Flood Forecasting and Warning Centre.

System Constraints

- Weather data must refresh automatically every 30 minutes
- Cyclone alert notifications must be sent even when app is closed
- Forecast accuracy data must be sourced from Bangladesh Meteorological Department
- Radar map must display rain intensity with color-coded overlay
- App must function with minimal data usage under 1MB per session

Project Duration	10 weeks	Estimated Budget	BDT 12,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 41 Job Portal and Recruitment Platform

Assigned To:	Fatima Jannat Dola	Student ID:	12411068
--------------	--------------------	-------------	----------

Project Overview

A job portal like BDJobs or Chakri.com is an online platform where employers post job vacancies and job seekers create profiles and apply with their CV. These platforms are the primary channel for formal job recruitment in Bangladesh, connecting fresh graduates with entry-level positions and experienced professionals with senior roles across industries like banking, IT, telecom, garments, and NGOs.

Your SRS should document the job seeker profile and CV builder system, the job posting and employer dashboard, the application tracking system for both sides, the job alert and recommendation engine, and the featured job and premium employer listing module. Cover the employer verification process and the application filtering tools including keyword-based CV screening. Also document the interview scheduling module and the recruiter messaging system for communicating with shortlisted candidates.

System Constraints

- CV builder must export formatted PDF with maximum 2 pages
- Job alert email must send within 1 hour of matching job posting
- Employer must be able to filter applications by education, experience, and location
- Application deadline must automatically close listings on expiry date
- Candidate contact info must be hidden until employer shortlists them

Project Duration	14 weeks	Estimated Budget	BDT 22,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 42 Telemedicine and Online Doctor Consultation App

Assigned To:	Muhammad Imtiaz Ahmed	Student ID:	12321076
--------------	-----------------------	-------------	----------

Project Overview

A telemedicine platform like Doctorola or Maya connects patients with licensed doctors for online consultations via video call, voice call, or text chat. Patients can describe symptoms, receive prescriptions, and get referred to specialists without traveling to a clinic. In Bangladesh, telemedicine became critical during COVID-19 and remains important for reaching patients in rural areas who lack access to quality healthcare facilities.

Your SRS must document the doctor registration and credential verification system, the appointment booking and queue management module, the video consultation interface with in-call prescription writing, the digital prescription and report management system, and the payment and subscription model requirements. Cover the follow-up appointment scheduling and the health record access system. Also document the emergency triage module that directs critical cases to hospital emergency services.

System Constraints

- Doctor credentials must be verified against BMDC registration database
- Video call quality must be maintained at minimum 480p on 3G connection
- Digital prescriptions must be signed with doctor's digital certificate
- Patient health records must be encrypted and patient-controlled
- Emergency triage cases must trigger immediate alert to on-call doctor

Project Duration	14 weeks	Estimated Budget	BDT 28,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 43

Smart City Waste Management System

Assigned To:	Md Touhidul Islam	Student ID:	12421106
--------------	-------------------	-------------	----------

Project Overview

A smart city waste management system allows municipal authorities and residents to track garbage collection schedules, report illegal dumping, monitor bin fill levels through IoT sensors, and optimize collection routes for waste trucks. City corporations in Dhaka and Chittagong are exploring digital solutions to improve waste collection efficiency and reduce urban sanitation problems. Citizens can use the app to report missed pickups and get notifications for their collection schedule.

Your SRS should document the citizen complaint and reporting module with location tagging, the waste collection schedule management system, the IoT bin sensor data integration requirements, the route optimization engine for collection vehicles, and the admin dashboard for monitoring city-wide waste operations. Cover the recycling category reporting system and the public cleanliness rating module. Also document the field worker mobile app for marking collection completion and reporting issues.

System Constraints

- Complaint reports must include geo-tagged photo as mandatory attachment
- Collection schedule must be viewable by ward and street level
- IoT sensor data must update bin fill levels every 4 hours
- Route optimization must reduce fuel usage by at least 15 percent versus manual routing
- Field worker app must function offline and sync when connectivity is restored

Project Duration	14 weeks	Estimated Budget	BDT 25,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 44 E-Government Citizen Service Portal

Assigned To:	Bappy Barai	Student ID:	12321106
--------------	-------------	-------------	----------

Project Overview

An e-Government portal like Bangladesh's e-services platform or the a2i initiative provides citizens with online access to government services including birth certificate applications, trade license renewal, tax filing, passport applications, land record searches, and utility connections. The goal is to reduce visits to government offices, eliminate middlemen, and make services accessible to all citizens regardless of location through a single unified digital platform.

Your SRS must document the citizen identity verification and NID integration system, the service catalog and application submission workflow, the document upload and verification module, the fee payment integration, and the application status tracking system with SMS notifications. Cover the government officer portal for processing and approving applications. Also document the grievance redressal system where citizens can escalate delayed or rejected service applications.

System Constraints

- NID verification must integrate with the Election Commission database
- All uploaded documents must be scanned for completeness before submission
- Application status must update within 24 hours of any processing action
- Fee payment must support mobile banking and online banking methods
- System must be accessible on feature phones via USSD for rural users

Project Duration	18 weeks	Estimated Budget	BDT 60,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.

Project 45

Reddit-Like Community Discussion Platform

Assigned To:	Miadam mahathir billah	Student ID:	12321027
--------------	------------------------	-------------	----------

Project Overview

Reddit is one of the world's largest online community platforms organized into thousands of topic-based communities called subreddits. Users post text, links, images, and videos which the community votes up or down to determine visibility. It is a major source of niche discussions, news commentary, technical help, entertainment, and community support. In Bangladesh, similar platforms and Facebook groups fill this role for local language community discussions.

Your SRS should document the community creation and moderation system, the post and comment submission workflow with voting mechanics, the karma and reputation system, the search and discovery module for finding communities and posts, and the Awards and premium membership module. Cover the moderator tools including post removal, user banning, and AutoModerator rule configuration. Also document the notification system for replies and mentions and the content reporting and admin review workflow.

System Constraints

- Vote counts must be displayed with fuzzy rounding to prevent manipulation
- New accounts must face posting restrictions for the first 7 days
- NSFW communities must require age verification before access
- AutoModerator must support keyword-based filtering and minimum karma rules
- Deleted posts must be hidden from users but retained for admin audit purposes

Project Duration	16 weeks	Estimated Budget	BDT 40,00,000
------------------	----------	------------------	---------------

SRS Deliverable: You must produce a complete SRS document following the IEEE 830 standard. It must include an Introduction (purpose, scope, definitions), Overall Description (product perspective, user classes, constraints), and Specific Requirements covering all functional modules, non-functional requirements, and external interface requirements.